

Majority Protocol Audit Summary

Prepared by Cyfrin

Version 2.0

Lead Auditors

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1 About Cyfrin

Cyfrin is a Web3 security company dedicated to bringing industry-leading protection and education to our partners and their projects. Our goal is to create a safe, reliable, and transparent environment for everyone in Web3 and DeFi. Learn more about us at cyfrin.io.

2 About This Document

This document is a public summary of the security audit Cyfrin performed for Majority on Majority Protocol. It records the repositories and commits reviewed, the timeline, auditors, in-scope files, and the number and final status of findings by severity. It closes with Cyfrin's attestation of the review's scope and final remediation status.

It is deliberately abbreviated: the technical detail of individual findings is not reproduced here. The complete audit report, version 2.0 dated January 27, 2026, was delivered to Majority, who retain it as the full record of the engagement. This summary was issued on October 29, 2026.

3 Disclaimer

The Cyfrin team makes every effort to find as many vulnerabilities in the code as possible in the given time but holds no responsibility for the findings in this document. A security audit by the team does not endorse the underlying business or product. The audit was time-boxed and the review of the code was solely on the security aspects of the in-scope code being audited. Unless the scope explicitly notes otherwise, the review is limited to the application layer and does not extend to the correctness of API/SDK/execution environment implementations, compilers, standard and third-party libraries, underlying operating systems and hardware.

4 Copyright and License

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5 Audit Overview

Over the course of 6 days, the Cyfrin team conducted an audit on the Majority Protocol code provided by Majority. The findings consist of 3 Critical, 5 High, 9 Medium & 7 Low severity issues with the remainder being informational and gas optimizations.

Summary

Project Name	Majority Protocol
Repository	engage-protocol
Commit	cca0cb3cedca...
Fix Commit	9b487e5a1560...
Audit Timeline	July 14th - July 21st, 2025
Methods	Manual Review

Issues Found

Severity	Found	Resolved	Acknowledged
Critical Risk	3	3	0
High Risk	5	5	0
Medium Risk	9	9	0
Low Risk	7	7	0
Informational	14	12	2
Gas Optimizations	8	8	0
Total	46	44	2

6 Audit Scope

The following files were in scope for this audit:

```
src/offchain/uma/SessionResultAsserter.sol
src/prompt/MajorityChoicePrompt.sol
src/prompt/SPBinaryPrompt.sol
src/prompt/TriviaChoicePrompt.sol
src/reward/FixedRanksReward.sol
src/reward/ProportionalToXPReward.sol
src/session/DefaultSession.sol
src/DepositManager.sol
src/QuestionManager.sol
src/Registry.sol
src/Roles.sol
src/SessionManager.sol
src/SessionManagerBase.sol
```

7 Risk Classification

	Impact: High	**Impact: Medium**	**Impact: Low**
Likelihood: High	Critical	High	Medium
Likelihood: Medium	High	Medium	Low
Likelihood: Low	Medium	Low	Low

8 Protocol Summary

Majority Protocol is a decentralized blockchain platform that enables creators to host trivia and opinion-based games where players compete for prizes. The protocol operates on a session-based model where game creators define entry fees, questions, reward distributions and game parameters. Players join games by paying entry fees in supported tokens (like USDC), which form a prize pool that is distributed to winners after deducting protocol and creator fees.

The core gameplay follows a commit-reveal pattern. Once a game starts, questions are revealed sequentially, and players submit hashed commitments of their answers during a reaction deadline period. After the deadline, players reveal their actual answers, which must match their commitments. The protocol supports multiple question types through different prompt strategies: TriviaChoice for traditional trivia with predetermined correct answers, MajorityChoice where the most popular answer wins, and SPBinary which uses the “Surprisingly Popular” algorithm for binary choices with probability predictions.

Game results are determined through integration with UMA’s Optimistic Oracle, which verifies winners and XP calculations. The protocol supports both fixed-rank rewards (predetermined percentages for 1st, 2nd, 3rd place, etc.) and proportional rewards based on XP earned. A referral system allocates 2% of entry fees to referrers. A Registry contract lists the session strategies, prompt strategies, reward strategies and payment tokens that can be used. The protocol also supports game cancellation with refunds, rescheduling, contestant limits and verification requirements for certain games.

9 Attestation

Cyfrin attests that it performed a security review of the Majority Protocol code provided by Majority.

The repositories, commits, review period and findings with their final status are in the Summary and Issues Found tables in Section 5, and the in-scope files in Section 6. This attestation is limited to the documented scope, audited commits and mitigations specifically reviewed by Cyfrin.

Cyfrin publishes this document and its SHA-256 hash at https://github.com/Cyfrin/cyfrin-audit-reports. To check a copy on macOS or Linux, run the following in the folder that contains it; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name. Before running it, check that the URL is Cyfrin's repository.

```
U=https://github.com/Cyfrin/cyfrin-audit-reports/raw/main/reports/sha256
N=2026-01-27-cyfrin-majority-protocol-v2.0.pdf
F=$N
H=$(curl -fsSL $U/$N.sha256 | cut -d' ' -f1)
echo "$H *$F" | shasum -a 256 -c
```

Cyfrin's full audit report with all findings provided to Majority has the SHA-256 hash and file name below. To check a copy of it, first check this document as above, then run the following in the folder that contains the full report; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name.

```
H=508670acb508bd6b2220fd380c58bed3bbd000b9ca6c220f94ffa366ada1ea32
F=2026-01-27-cyfrin-majority-protocol-v2.0-full.pdf
echo "$H *$F" | shasum -a 256 -c
```

On Linux, `sha256sum -c` can be used in place of `shasum -a 256 -c`. On Windows, run `Get-FileHash <file> -Algorithm SHA256` in PowerShell and compare the result with the hash.